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Clinical Outcome and Therapeutic Impact on Neuroendocrine Neoplasms of the Breast: A National Cancer Database Study

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Abstract

Purpose: Neuroendocrine neoplasms (NENs) of the breast are rare and not well-studied. NEN are subcategorized as well-differentiated neuroendocrine tumor (NET) and poorly differentiated neuroendocrine carcinoma (NEC). The objectives of the current study were to review the clinicopathologic features of NENs, therapeutic efficacy of current systemic therapy and clinical outcomes of NEN of the breast.

Methods: Between 2004 and 2015, 420 NET, 205 NEC, 146 Adenocarcinoma with NE differentiation (ACNED) and 1,479,520 of invasive carcinoma, not otherwise specified (IC-NOS) of the breast were identified in the National Caner Database. Overall survival was compared among groups using Kaplan-Meier method and Log-rank test. Multivariate analyses were performed to identify prognostic factors.

Results: After adjusting for other prognostic factors, both NET and NEC of the breast showed significantly worse OS than IC-NOS (HR (95% CI) = 1.41 (1.17, 1.72), P=0.005 and HR (95% CI) = 2.11 (1.67, 2.67), P<0.001, respectively). Both NET and NEC benefited from endocrine therapy if the tumors were hormonal receptor positive (median OS for treated with vs without: 125 vs 57 months in NET, not reached vs 29 months in NEC). NEC also benefited from chemotherapy (median OS for treated with vs without: 42 vs 34 months), but not NET.

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Conclusion: NEN is a unique pathologic and clinical entity, which has worse clinical outcome compared to IC-NOS of the breast. Current therapeutics used in the treatment of IC-NOS improve, but do not fully mitigate, the poorer prognosis of NEN patients. More effective therapy for patients with this unique tumor type are needed.

Keywords

Carcinoma; Neuroendocrine tumor; Neuroendocrine Neoplasms; Breast; Prognosis

INTRODUCTION

Neuroendocrine neoplasms (NEN) of the breast are a group of evolving and under-appreciated malignant neoplasms. It was first proposed as a distinct entity by WHO classification of the breast tumors in 2003 [1]. It was then further defined as 3 subcategories -well-differentiated neuroendocrine tumor, poorly differentiated neuroendocrine carcinoma and adenocarcinoma with neuroendocrine differentiation- in the subsequent edition of WHO classification in 2012 [2]. In the most current edition (2019), the diagnostic criteria and terminology adopted a uniform classification framework for neuroendocrine neoplasms in all the organ systems by the International Agency for Research on Cancer (IARC) and World Health Organization (WHO) expert panel [3, 4]. The neuroendocrine neoplasms are comprised of two categories, the well-differentiated neuroendocrine tumor (NET) and the poorly differentiated neuroendocrine carcinoma (NEC). The well-differentiated neuroendocrine tumor includes grade 1- grade 3 neuroendocrine tumors. The poorly differentiated neuroendocrine carcinoma includes small cell and large cell neuroendocrine carcinoma. In addition, the adenocarcinoma with neuroendocrine differentiation (ACNED) was dropped out in the neuroendocrine neoplasm category due to its ambiguity without clear diagnostic criteria and clinical significance.

Due to its rarity, there is no consensus as to how NENs should be treated, and if there is different clinical prognosis from invasive carcinoma not otherwise specified (IC-NOS). Particularly, the well-differentiated NETs of the breast are always treated as the same as invasive carcinoma of the breast. So far, there is no data on whether the similar treatment as invasive mammary carcinoma is effective for neuroendocrine neoplasm of the breast. National Cancer Database (NCDB) is the largest cancer registry in US, which not only contains the largest collection of the patients, it also provided the treatment information for the patients who were treated in the accredited cancer programs in US.

The objectives of the current study are to 1) review the clinical pathologic features and clinical outcomes 2) to estimate therapeutic efficacy of current therapy in neuroendocrine neoplasms of the breast, in contrast to IC-NOS using the National Cancer Database (NCDB).

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MATERIALS AND METHODS

Patient Population

The NCDB is a hospital-based registry sponsored by the Commission on Cancer of the American College of Surgeons and the American Cancer Society. It includes data of approximately 70% of the newly diagnosed cancer cases from > 1500 commission-accredited cancer programs [5]. Data registry contains patient demographic characteristics, cancer stage, cancer histologic characteristics, type of treatment administrated and time to death or censoring. Because patient identifiers are not available to database users, this study was exempted for institutional review board review and approval. 2019 NCDB participant user file was used for this study.

Since NEN of the breast was first established by WHO in 2003, cases were selected from NCDB starting from 2004 and ending in 2015 in order to have sufficient follow up for survival analyses.

Study cases were extracted from the NCDB based on cancer histology codes of the third edition of the International Classifications of Diseases for Oncology (ICD-O-3). Invasive breast carcinoma, not otherwise specified (NOS) (C500–509/3), invasive well-differentiated neuroendocrine tumor (NET) (8240/3, 8246/3, 8249/3), poorly differentiated neuroendocrine carcinoma (NEC), including small cell carcinoma (8041–8045/3) and large cell neuroendocrine carcinoma (8013/3) were retrieved from the database for analyses (Figure. 1). For comparison, 136 cases of adenocarcinoma with neuroendocrine differentiation (ACNED) (8574/3) were also retrieved from the database (Figure 1).

Statistical Analyses

Overall survival (OS) was measured from the time of diagnosis to the time of death. OS time for the surviving patients was right censored at the time of last contact. The distribution of OS was estimated by the Kaplan-Meier method [6]. Log-rank test [7] was performed to test the difference in survival between groups. Regression analyses of survival data based on the Cox proportional hazards (PH) model [8] were conducted on OS in multivariate setting.

All tests were two-sided. P-values less than 0.05 were considered statistically significant. All analyses were conducted using SAS (version 9.4, Cary, NC) and S-plus (version 8.04, TIBCO Software Inc., Palo Alto, CA) statistical software.

RESULTS

Incidence

From 2004 to 2015, a total of 1,480,291 patients with invasive breast cancers were registered in NCDB. Among them, 420 NET, 205 NEC and 146 ACNED were identified. Patient selection is illustrated in Figure 1.

Due to significant imbalance between the number of patients with IC-NOS and NENs, 3,855 patients, five times the number of neuroendocrine neoplasm (n=771) were randomly

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selected from the IC-NOS pool, and patient demographic, clinicopathologic and treatment characteristics were compared (Table 1).

The proportion of patients diagnosed with any type of NEN steadily and significantly increased between 2004 to 2015 compared to IC-NOS ($p < 0.01$) (Table 1). The increase was particularly notable for NET; of the 422 NET cases in this cohort, only 11 cases (2.6% of total) were diagnosed in 2004 as compared to 56 cases (13% of total) in 2015.

Clinicopathologic characteristics

The mean age at diagnosis for patients with well differentiated NET was 63 years (range 26 – 90, median 63), and for poorly differentiated NEC 61 years (range 28 – 90, median 60), as compared to an average age of 60 years (range 21 – 90, median 60) for patients with IC-NOS ($p < 0.01$) (Table 1).

At the time of the diagnosis, patients with NET or NEC tended to present at higher clinical stage with >10% presenting as Stage IV compared to 4.4% of IC-NOS presenting with Stage IV disease ($p <0.05$). NET and NEC tumors were also more likely to be assigned a higher histologic grade, were less likely to be hormone receptor (HR) positive, and less likely to be HER2 positive as compared to IC-NOS (all $p < 0.01$) (Table 1). No significant differences were noted between the groups with regards race distribution, treating facility location and treating facility type.

Treatment modalities

Due to high proportion of patients with NET (12%) and NEC (18%) presented as stage IV disease, compared to 4.4% of IC-NOS patients, surgery was omitted as part of the treatment plan in 23.4% of NET and 19.5% of NEC patients compared to 13.2% of IC-NOS patients ($p < 0.01$) (Table 1).

Consistent with lower rates of HR+ positive disease, patients with NEC and NET were also less likely to receive endocrine therapy (19.6% and 51% respectively compared to 64.3% of IC-NOS, $p < 0.01$). In contrast, patients with NEC and NET were more likely to receive chemotherapy compared to patients with IC-NOS (78.9%, 53.4% and 47.3% respectively, $p < 0.01$) (Table 1). There was no difference in utilization of radiation therapy among the groups (Table 1).

Overall survival

The median follow-up was 69 months and median overall survival (OS) was 161.8 months for the entire group. Patients who had NENs had inferior OS than those who had IC-NOS; median OS were 105.5, 38.3 and 163.5 months for well differentiated NET, poorly differentiated NEC and IC-NOS, respectively, $p <0.01$ (Figure 2). ACNED, an old entity currently dropped out from the WHO classification, showed median OS of 150.1 month, not significantly different from the OS of IC-NOS group, $p= 0.99$ (Figure 2).

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Multivariate analysis

Multivariate analysis of OS in the entire cohort shows that tumor type (p <0.01), age (p <0.01), insurance status (p <0.01), Charlson/Deyo comorbidity score (p <0.01), clinical stage (p <.001), tumor grade (p <0.01), chemotherapy (p <0.01), surgery (p <0.01), radiation therapy (p <0.01) and hormone therapy (p <0.01) were significantly associated with OS. Specifically, patients who had well differentiated NET had inferior OS than those who had IC-NOS (HR (95% CI) = 1.4 (1.17, 1.72), p <0.01); patients who had poorly differentiated NEC also had inferior OS than those who had IC-NOS (HR (95% CI) = 2.12 (1.67, 2.67), p <0 .01). However, the difference of OS between the patients who had ACNED and IC-NOS was not statistically significant (p = 1.00) (Table 2). Increasing age, not insured or public insurance, higher Charlson-Deyo comorbidity index, higher clinical TNM stage, high histologic grade, no chemo, no radiation and no endocrine therapy received (P< 0.01) were associated with inferior OS (Table 2).

Impact of systemic therapy on OS

We next analyzed the efficacy of systemic therapy in each of the tumor group to evaluate if the treatments had differential effects in different tumor types.

As shown in Figure 3, both IC-NOS and poorly differentiated NEC benefited from chemotherapy with cohorts treated with chemotherapy doing significantly better than those treated without chemotherapy (p = 0.02 and 0.04 for IC-NOS and NEC respectively). Median survival among IC-NOS patients without chemotherapy was 106 months and was not reached for those who did receive chemotherapy. Median survival for NEC patients who did not receive chemotherapy was 34 months and this improved to 42 months with receipt of chemotherapy (Table 3). There was no significant difference in OS with and without chemotherapy for patients with NET (Figure 3, Table 3).

As shown in Figure 4 and Table 4, HR+ patients in all three tumor groups were benefited from receipt of endocrine therapy. Patients in each of the tumor groups had better OS if they received endocrine than those who did not. The median, 5-year and 10-year OS are summarized in Tables 4.

DISCUSSION

Given the rarity of NENs, there is a paucity of data on incidence, pathologic features, treatment impact and survival. Compared to our previous study from SEER database, which records cancer incidence in 27.8% of the US population, the current study takes advantage of larger database from NCDB, which collects approximately 70% of the newly diagnosed cancer cases in US. Both studies demonstrate that the incidence of NEN of the breast has significantly increased from 2004 to 2015, particularly for NET. However, it is unclear whether the incidence has truly increased or whether there has just been increased recognition of this rare entity by pathologists.

In this large database study, we demonstrate that NENs are overall less favorable compared to IC-NOS, but also show heterogeneity with different outcomes and different treatment

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effects based on the specific subtype of NEN. These data are in line with other recent series demonstrating the distinct biology of NEN tumors.

One key area of controversy has been the prognostic relevance of NEN diagnosis. Results from some institutional studies have shown worse clinical outcome [9–12], while others show similar or better outcomes compared to invasive mammary carcinoma, NOS [13, 14]. Our series demonstrate that NEN of the breast carries worse clinical outcome compared to invasive mammary carcinoma, NOS. Our study result corroborates a recent NCDB study [15] and SEER (surveillance, epidemiology, and end results) studies [16–18], and thus underscores the importance of recognizing breast NEN as a distinct clinicopathologic entity as compared to IC-NOS.

Additionally, as shown by other studies [19, 20], our study also demonstrated that there are no overall survival differences between the ACNED and IC-NOS. Together with its ambiguity of diagnostic criteria, ACNED is no longer considered a separate entity within the neuroendocrine neoplasm category by WHO. Instead, it is considered as IC-NOS with focal neuroendocrine differentiation.

Our study also evaluated the impact of therapeutic modalities on mammary NEN. Currently, there are no consensus treatment recommendation for mammary NEN. Most of mammary NENs are currently treated as IC- NOS. Patients are subject to surgical resection, endocrine therapy, chemotherapy and radiation therapy based on stage and hormonal status. Our data demonstrate that HR positive NENs benefit from ET. Our data show however that chemotherapy has different therapeutic effects on well differentiated NET and poorly-differentiated NEC. This study showed that patients with poorly differentiated NEC benefited from chemotherapy, but not patients with well differentiated NET. A recent study showed that mammary NEN patients who received chemotherapy with anthracycline/taxane and or taxane-containing regimens had better OS and DFS than those without those regimens [21]. Therefore, the chemotherapeutic regimens may be optimized for better treatment of mammary NET. Nonetheless, despite the therapeutic benefit of chemotherapy for NEC patients, the receipt of chemotherapy did not fully mitigate the overall negative prognosis of this subtype of breast cancer.

The strengths of this study include its large cohort size representing one of the largest studies on primary NEN of the breast and captures data from a diverse cross section of the US. Given the retrospective nature of the study, several limitations are noted. As with all observational studies, treatment biases cannot be fully adjusted for and could impact the OS findings reported herein. The details of chemotherapy regimens are not available in NCDB therefore precluding assessment of efficacy of individual regiments. In addition, the HER2 status was not available prior 2010 in the NCBD, thus limiting ability to look specifically at this subtype. Although only a small percentage of mammary NEN are HER2 positive, it will be important to evaluate whether HER2 targeted therapy is effective in HER2-positve mammary NENs.

In conclusion, mammary NEN including well-differentiated NET and poorly differentiated NEC is a distinct type of malignant neoplasms in the breast, which carries worse prognosis

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than invasive mammary carcinoma, NOS. Chemotherapy appears to improve OS in poorly differentiated NEC and HR+ mammary NEN appears to benefit from endocrine therapy. However, overall current systemic regimens do not fully mitigate the poor prognosis overall of NENs and studies testing additional therapeutic interventions are needed.

Funding Statement

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Abbreviations

ACNED	Adenocarcinoma with neuroendocrine differentiation
NCDB	National Cancer Database
NEN	Neuroendocrine neoplasm
NE	Neuroendocrine differentiation
NEC	Neuroendocrine carcinoma
IARC	International Agency for Research on Cancer
RFS	Recurrence-free survival
OS	Overall survival
SEER	Surveillance, epidemiology and end results
IC-NOS	Invasive carcinoma, not otherwise specified
ER	Estrogen receptor
HER2	Human Epidermal Growth Factor Receptor type 2
CI	Confidential interval
PR	Progesterone receptor
HR	Hormonal receptor

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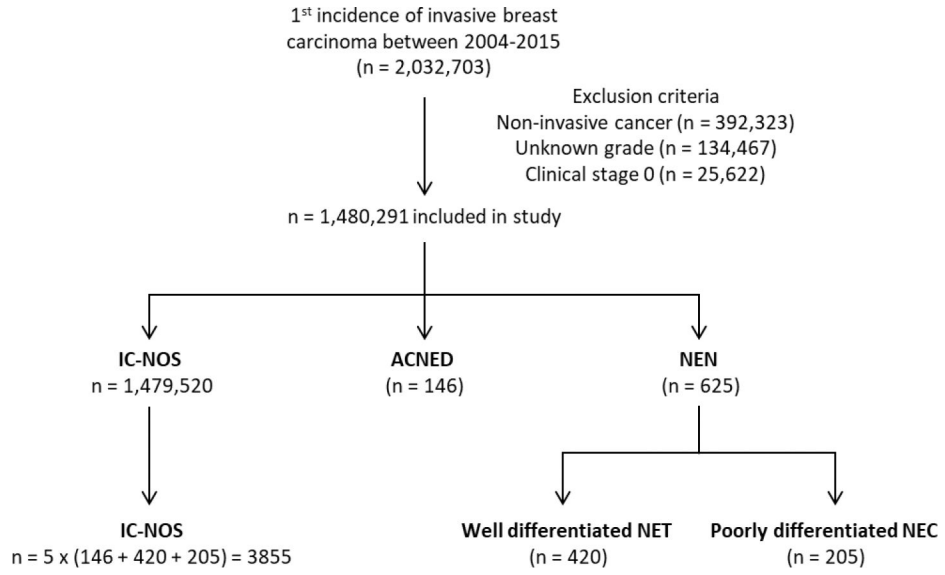

Figure 1.
Flow chart for patient selection

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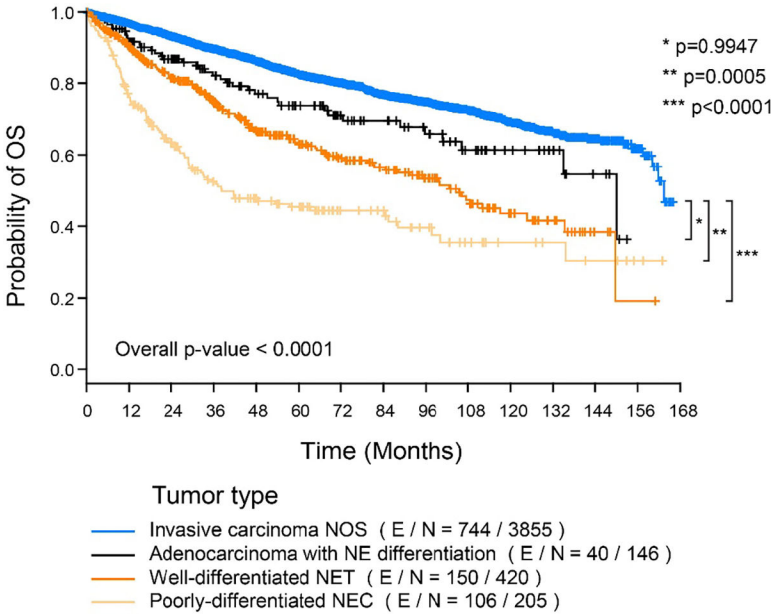

Figure 2.

Kaplan-Meier analysis of overall survival (OS) of patients with invasive carcinoma, NOS, and neuroendocrine neoplasms including adenocarcinoma with neuroendocrine differentiation, well differentiated neuroendocrine tumor (NET) and poorly differentiated neuroendocrine carcinoma (NEC) in the National Cancer Database from 2004 to 2015. Well differentiated NET and poorly differentiated NEC have significant worse OS compared to invasive carcinoma, NOS in nonmatched cohort (*p*-Value = 0.0005 and < 0.000, respectively) (E is the event of death and N is the total number within the defined group [E/N]). However, OS between the patients who had adenocarcinoma with NE differentiation and IC-NOS was not statistically significant (*p*-Value = 0.9947)

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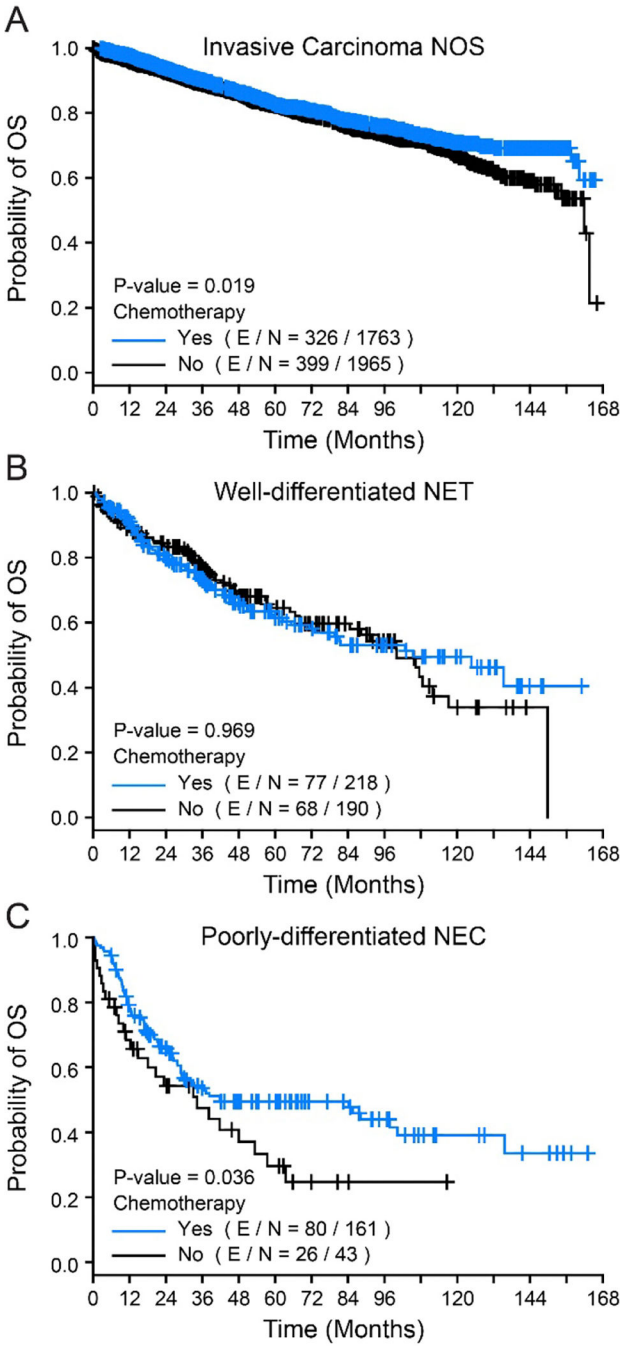

Figure 3.

Chemotherapy effect on overall survival (OS) of patients with invasive carcinoma, NOS, well-differentiated neuroendocrine tumor (NET) and poorly-differentiated neuroendocrine carcinoma (NEC). Kaplan–Meier curves demonstrate that patients of invasive carcinoma, NOS and poorly-differentiated NEC, who received chemotherapy had better OS than those who did not (**A** and **C**). Patients with well-differentiated NET have no survival benefit whether they received chemotherapy or not (**B**). (E is the event of death and N is the total number within the defined group [E/N])

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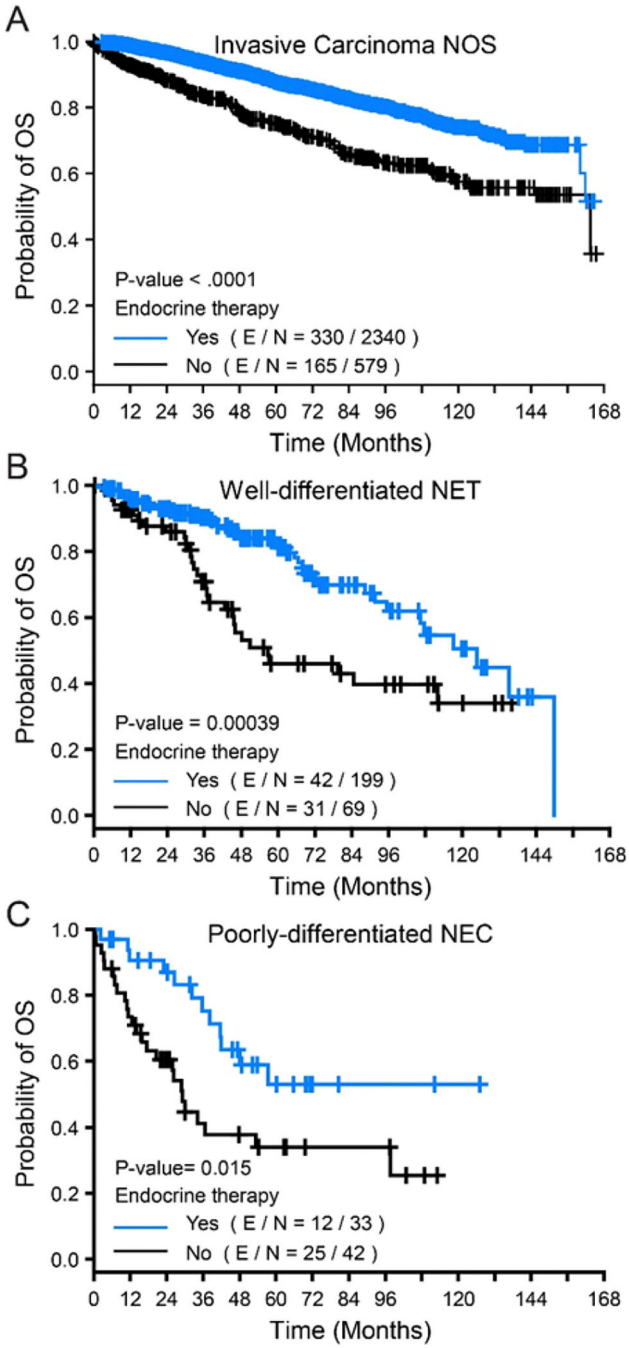

Figure 4.

Endocrine therapy on overall survival (OS) of patients with invasive carcinoma, NOS, well-differentiated neuroendocrine tumor (NET) and poorly-differentiated neuroendocrine carcinoma (NEC). Kaplan–Meier curves demonstrate that patients of invasive carcinoma, NOS (**A**), well-differentiated NET (**B**) and poorly-differentiated NEC (**C**), who received endocrine therapy had better OS than those who did not. (E is the event of death and N is the total number within the defined group [E/N])

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Table 1

Patient Demographic, Clinical, and Treatment Characteristics: National Cancer Database, 2004 to 2015 among patients of NEC, NET, ACNED, IC-NOS (n = 4626)

Covariate	Levels	PDNEC (*n* = 205)	WDNET (*n* = 422)	ACNED (*n* = 147)	IC-NOS (*n* = 3855)	*p*-value
Age: Mean		60.81	63.21	62.99	60.11	< 0.0001
Median (range)		60 (28– 90)	63 (20–90)	62.5 (25–90)	60 (21–90)	
Year of diagnosis	2004	15 (7.3%)	11 (2.6%)	7 (4.8%)	265 (6.9%)	0.0000
	2005	5 (2.4%)	24 (5.7%)	17 (11.6%)	259 (6.7%)	
	2006	11 (5.4%)	38 (9%)	11 (7.5%)	291 (7.5%)	
	2007	17 (8.3%)	26 (6.2%)	7 (4.8%)	314 (8.1%)	
	2008	10 (4.9%)	33 (7.9%)	16 (11%)	324 (8.4%)	
	2009	16 (7.8%)	22 (5.2%)	16 (11%)	325 (8.4%)	
	2010	20 (9.8%)	22 (5.2%)	8 (5.5%)	330 (8.6%)	
	2011	24 (11.7%)	32 (7.6%)	17 (11.6%)	338 (8.8%)	
	2012	26 (12.7%)	42 (10%)	6 (4.1%)	319 (8.3%)	
	2013	19 (9.3%)	63 (15%)	7 (4.8%)	353 (9.2%)	
	2014	15 (7.3%)	51 (12.1%)	14 (9.6%)	352 (9.1%)	
	2015	27 (13.2%)	56 (13.3%)	20 (13.7%)	385 (10%)	
Race	Black	32 (15.7%)	49 (11.7%)	13 (9%)	436 (11.4%)	0.1666
	Other	10 (4.9%)	9 (2.2%)	7 (4.9%)	159 (4.2%)	
	White	162 (79.4%)	360 (86.1%)	124 (86.1%)	3223 (84.4%)	
DCCI total best	0	161 (78.5%)	332 (79%)	118 (80.8%)	3296 (85.5%)	0.0022
	1	32 (15.6%)	69 (16.4%)	25 (17.1%)	454 (11.8%)	
	2	10 (4.9%)	13 (3.1%)	2 (1.4%)	75 (1.9%)	
	3	2 (1%)	6 (1.4%)	1 (0.7%)	30 (0.8%)	
Income level	< $30,000	16 (7.8%)	52 (12.4%)	18 (12.3%)	405 (10.5%)	0.6818
	$30,000-$34,999	41 (20%)	70 (16.7%)	23 (15.8%)	599 (15.5%)	
	$35,000-$45,999	54 (26.3%)	116 (27.6%)	37 (25.3%)	995 (25.8%)	
	$46,000 +	87 (42.4%)	169 (40.2%)	63 (43.2%)	1746 (45.3%)	

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Covariate	Levels	PDNEC (*n* = 205)	WDNET (*n* = 422)	ACNED (*n* = 147)	IC-NOS (*n* = 3855)	*p*-value
	Unknown	7 (3.4%)	13 (3.1%)	5 (3.4%)	110 (2.9%)	
Insurance status	Public insurance	89 (43.4%)	211 (50.2%)	67 (45.9%)	1634 (42.4%)	0.0001
	Private insurance	101 (49.3%)	184 (43.8%)	74 (50.7%)	2053 (53.3%)	
	Not insured	13 (6.3%)	9 (2.1%)	3 (2.1%)	87 (2.3%)	
	Unknown	2 (1%)	16 (3.8%)	2 (1.4%)	81 (2.1%)	
Home location	Metro	174 (84.9%)	345 (82.1%)	129 (88.4%)	3192 (82.8%)	0.5709
	Urban	25 (12.2%)	58 (13.8%)	13 (8.9%)	500 (13%)	
	Rural	4 (2%)	6 (1.4%)	3 (2.1%)	59 (1.5%)	
	Unknown	2 (1%)	11 (2.6%)	1 (0.7%)	104 (2.7%)	
Treatment facility	Academic/research program	58 (28.3%)	137 (32.6%)	56 (38.4%)	1095 (28.4%)	0.1563
	Community cancer program	50 (24.4%)	89 (21.2%)	27 (18.5%)	890 (23.1%)	
	Comprehensive community cancer program	82 (40%)	177 (42.1%)	56 (38.4%)	1639 (42.5%)	
	Unknown	15 (7.3%)	17 (4%)	7 (4.8%)	231 (6%)	
Facility location	Midwest	42 (20.5%)	104 (24.8%)	36 (24.7%)	900 (23.3%)	0.6248
	Northeast	45 (22%)	84 (20%)	30 (20.5%)	818 (21.2%)	
	South	76 (37.1%)	134 (31.9%)	49 (33.6%)	1290 (33.5%)	
	Unknown	15 (7.3%)	17 (4%)	7 (4.8%)	231 (6%)	
	West	27 (13.2%)	81 (19.3%)	24 (16.4%)	616 (16%)	
TNM clinical stage	1	27 (13.2%)	129 (30.7%)	35 (24%)	1587 (41.2%)	0.0000
	2	73 (35.6%)	120 (28.6%)	37 (25.3%)	926 (24%)	
	3	29 (14.1%)	50 (11.9%)	11 (7.5%)	233 (6%)	
	4	37 (18%)	51 (12.1%)	20 (13.7%)	168 (4.4%)	
	Unknown	39 (19%)	70 (16.7%)	43 (29.5%)	941 (24.4%)	
Grade	1	2 (1%)	72 (17.1%)	21 (14.4%)	842 (21.8%)	0.0000
	2	17 (8.3%)	131 (31.2%)	58 (39.7%)	1663 (43.1%)	
	3	154 (75.1%)	200 (47.6%)	66 (45.2%)	1330 (34.5%)	
	4	32 (15.6%)	17 (4%)	1 (0.7%)	20 (0.5%)	
Hormone receptor	Negative	104 (10.7%)	119 (12.3%)	23 (2.4%)	724 (74.6%)	< 0.0001
	Positive	83 (2.4%)	281 (8%)	117 (3.3%)	3043 (86.4%)	

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Covariate	Levels	PDNEC (*n* = 205)	WDNET (*n* = 422)	ACNED (*n* = 147)	IC-NOS (*n* = 3855)	*p*-value
	Unknown	18 (13.6%)	20 (15.2%)	6 (4.5%)	88 (66.7%)	
HER2*	Negative	114 (55.6%)	235 (56%)	67 (45.9%)	1707 (44.3%)	0.0000
	Positive	1 (0.5%)	8 (1.9%)	3 (2.1%)	289 (7.5%)	
	Unknown	90 (43.9%)	177 (42.1%)	76 (52.1%)	1859 (48.2%)	
Chemotherapy	No	43 (21.1%)	190 (46.6%)	69 (48.6%)	1965 (52.7%)	0.0000
	Yes	161 (78.9%)	218 (53.4%)	73 (51.4%)	1763 (47.3%)	
Surgery	No	40 (19.5%)	98 (23.4%)	21 (14.4%)	507 (13.2%)	0.0000
	Yes	165 (80.5%)	321 (76.6%)	125 (85.6%)	3346 (86.8%)	
Radiation therapy	No	84 (41.2%)	164 (39.6%)	56 (38.4%)	1532 (40.1%)	0.9559
	Yes	120 (58.8%)	250 (60.4%)	90 (61.6%)	2287 (59.9%)	
Endocrine therapy	No	152 (80.4%)	197 (49%)	48 (34.3%)	1317 (35.7%)	0.0000
	Yes	37 (19.6%)	205 (51%)	92 (65.7%)	2373 (64.3%)	

*HER2 status is only available since 2010 in NCDB

PDNEC Poorly-differentiated neuroendocrine carcinoma, *WDNET* Well-differentiated neuroendocrine tumor, *ACNED* adenocarcinoma with neuroendocrine differentiation, *IC-NOS* Invasive carcinoma, not otherwise specified, *DCCI* Charlson-Deyo comorbidity index

Conflict of Interest Statement

The authors do not have any conflicts of interest to declare.

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Table 2

Multivariate analysis of prognostic factors for overall survival

Variable		Hazard ratio	95% Hazard Ratio	*p*-value
Tumor type	IC-NOS (ref)			< 0.0001
	ACNED	1.00	0.71–1.40	
	Well-differentiated NET	1.41	1.17–1.72	
	Poorly-differentiated NEC	2.11	1.67–2.67	
Age	Per year increase	1.04	1.03–1.04	< 0.0001
Insurance status	Private insurance (ref)			0.0001
	Public insurance	1.34	1.14–1.58	
	Not insured	1.39	0.93–2.07	
DCCI	0 (ref)			< 0.0001
	1	1.28	1.08–1.53	
	2	2.18	1.62–2.95	
	3	2.66	1.70–4.17	
TNM clinical stage	1 (ref)			< 0.0001
	2	2.21	1.80–2.73	
	3	3.99	3.09–5.15	
	4	13.17	10.39–16.71	
	Unknown	1.66	1.36–2.03	
Grade	1 (ref)			< 0.0001
	2	1.36	1.00–1.69	
	3	1.75	1.40–2.18	
	4	2.55	1.71–3.81	
Surgery	No (ref)			< 0.0001
	Yes	0.67	0.56–0.81	
Endocrine therapy	No (ref)			< 0.0001
	Yes	0.53	0.46–0.61	
Chemotherapy	No (ref)			0.0024
	Yes	0.78	0.66–0.92	

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Variable		Hazard ratio	95% Hazard Ratio	*p*-value
Radiation therapy	No (ref)			< 0.0001
	Yes	0.74	0.65–0.84	

IC-NOS Invasive carcinoma, not otherwise specified, *ACNED* Adenocarcinoma with neuroendocrine differentiation, *NET* Neuroendocrine tumor, *NEC* Neuroendocrine carcinoma, *DCCI* Charlson-Deyo comorbidity index

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Table 3

Summary statistics of OS by chemotherapy

Tumor type	Chemotherapy	Event/Total	Median OS (months)	5-year OS rate	10-year OS rate
Invasive carcinoma—NOS	No	399/1965	161.81 (151.98, NR)	0.82 (0.80, 0.84)	0.67 (0.63, 0.70)
	Yes	326/1763	NR (160.16, NR)	0.83 (0.80, 0.84)	0.71 (0.68, 0.74)
Poorly-differentiated NEC	No	26/43	34.20 (12.29, 53.26)	0.30 (0.15, 0.46)	NR
	Yes	80/161	41.95 (28.94, 100.21)	0.49 (0.41, 0.57)	0.39 (0.29, 0.49)
Well-differentiated NET	No	68/190	100.04 (84.57, 117.13)	0.65 (0.56, 0.72)	0.34 (0.21, 0.47)
	Yes	77/218	105.46 (71.29, NR)	0.61 (0.53, 0.69)	0.5 (0.40, 0.59)

NR not reached

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Table 4

Summary statistics of OS by endocrine therapy among the HR positive patients

Tumor type	Endocrine therapy	Event/Total	Median OS (months)	5-year OS rate	10-year OS rate
Invasive carcinoma—NOS	No	165/579	163.45 (124.02, NR)	0.75 (0.71, 0.79)	0.57 (0.51, 0.63)
	Yes	330/2340	NR (160.16, NR)	0.88 (0.86, 0.89)	0.74 (0.71, 0.77)
Poorly-differentiated NEC	No	25/42	29.11 (15.74, 97.64)	0.34 (0.19, 0.50)	NR
	Yes	12/33	NR (41.66, NR)	0.53 (0.31, 0.71)	0.53 (0.31, 0.71)
Well-differentiated NET	No	31/69	56.71 (42.68, NR)	0.46 (0.32, 0.59)	0.34 (0.19, 0.50)
	Yes	42/199	124.68 (95.44, 149.75)	0.81 (0.73, 0.87)	0.50 (0.34, 0.65)

NR not reached